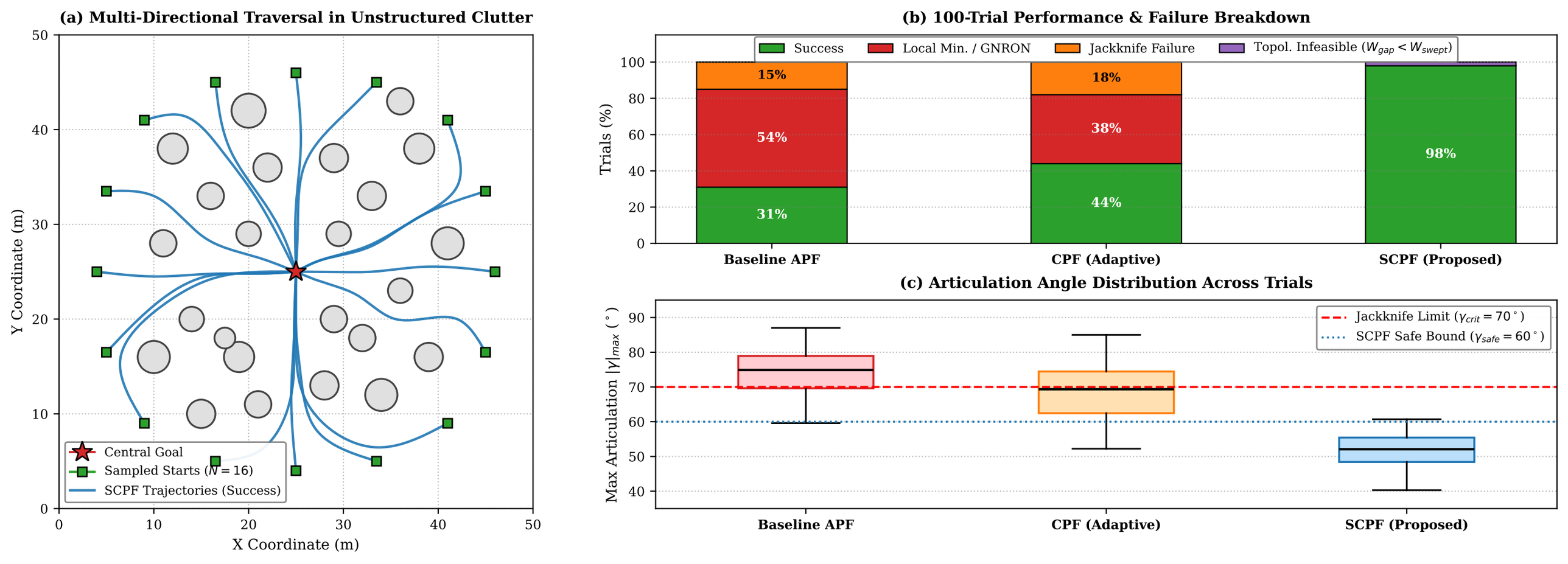

(a) Multi-Directional Traversal in Unstructured Clutter
Central Goal
Sampled Starts (N = 16)
SCPF Trajectories (Success)
X Coordinate (m)
Y Coordinate (m)
(b) 100-Trial Performance & Failure Breakdown
Success
Local Min. / GNRON
Jackknife Failure
Topol. Infeasible ($W_{gap} < W_{swept}$)
Trials (%)
15%
54%
31%
18%
38%
44%
98%
Baseline APF
CPF (Adaptive)
SCPF (Proposed)
(c) Articulation Angle Distribution Across Trials
Jackknife Limit ($\gamma_{crit} = 70^\circ$)
SCPF Safe Bound ($\gamma_{safe} = 60^\circ$)
Max Articulation $|\gamma|_{max}$ ($^\circ$)
Baseline APF
CPF (Adaptive)
SCPF (Proposed)